usv_hydro

Fiziksel Modeller ve Formuller

Teknik Referans Raporu

Proje	Maritime Autonomous Systems Simulator
Paket	`usv_hydro` v0.1.0
Platform	ROS 2 Jazzy + Gazebo Sim 8
Tarih	Nisan 2026
Durum	Faz 1 Tamamlandi – 21/21 test gecti

Bu rapor; her alt modelin girdi–cikti tanimlarini, uygulanan fiziksel formuller ve parametrelerini, modeller arasi veri akisini ayrintili bicimde aciklamaktadir.

Kaynak dosyalar:

`HydroTypes.cc`, `HydroCellSystem.cc`, `BuoyancyModel.cc`, `DragModel.cc`, `HydroIntegrator.cc`, `EnvironmentModel.cc`

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1 Giris ve Sistem Mimarisi

1.1 Genel Bakis

usv_hydro, Gazebo Sim 8 uzerinde calisan bir C++ fizik plugin'idir. Her simulasyon adiminda (ISystemPreUpdate) Gazebo'dan govde kinematik verileri alinir, alti fizik alt-modeli hesaplanir ve elde edilen kuvvet/moment Gazebo fizik motoruna enjekte edilir.

1.2 Sistem Girdisi – Gazebo'dan Gelen Veriler

Table 1: Her adimda Gazebo'dan okunan girdi degiskenleri

Degisken	Tur	Aciklama
$\mathbf{p}_W$	$\mathbb{R}^3$	Dunya cerçevesinde govde konumu (m)
$\mathbf{q}_W$	quaternion	Dunya cerçevesine gore donus
$\mathbf{v}^W$	$\mathbb{R}^3$	Dunya cerçevesinde dogrusal hiz (m/s)
$\boldsymbol{\omega}^W$	$\mathbb{R}^3$	Dunya cerçevesinde acisal hiz (rad/s)
Δt	skaler	Simulasyon adim suresi (s)
t	skaler	Toplam simulasyon zamani (s)

Plugin bu degerlerden body-frame hiz ve ivme vektorlerini turetir:

$$\boldsymbol{\nu} = \begin{bmatrix} \mathbf{R}^T \mathbf{v}^W \\ \mathbf{R}^T \boldsymbol{\omega}^W \end{bmatrix} \in \mathbb{R}^6 \tag{1}$$

$$\dot{\boldsymbol{\nu}} \approx \frac{\boldsymbol{\nu}(t) - \boldsymbol{\nu}(t - \Delta t)}{\Delta t} \in \mathbb{R}^6 \tag{2}$$

burada $\mathbf{R}$ = dunya→govde donum matrisi ($\mathbf{q}_W$'dan).

DOF sirasi tum modellerde ortaktir:

[0=surge, 1=sway, 2=heave, 3=roll, 4=pitch, 5=yaw]

1.3 Toplam Kuvvet Bilesimi

Her adimda hesaplanan kuvvetler Gazebo'ya AddWorldWrench ile eklenir:

$$\boxed{\mathbf{F}_{\text{toplam}} = \mathbf{F}_{\text{kald}} + \mathbf{F}_{\text{direnc}} + \mathbf{F}_{\text{rijitlik}} + \mathbf{F}_{\text{radyasyon}} + \mathbf{F}_{\text{uyarim}}} \tag{3}$$

1.4 Veri Akis Semasi

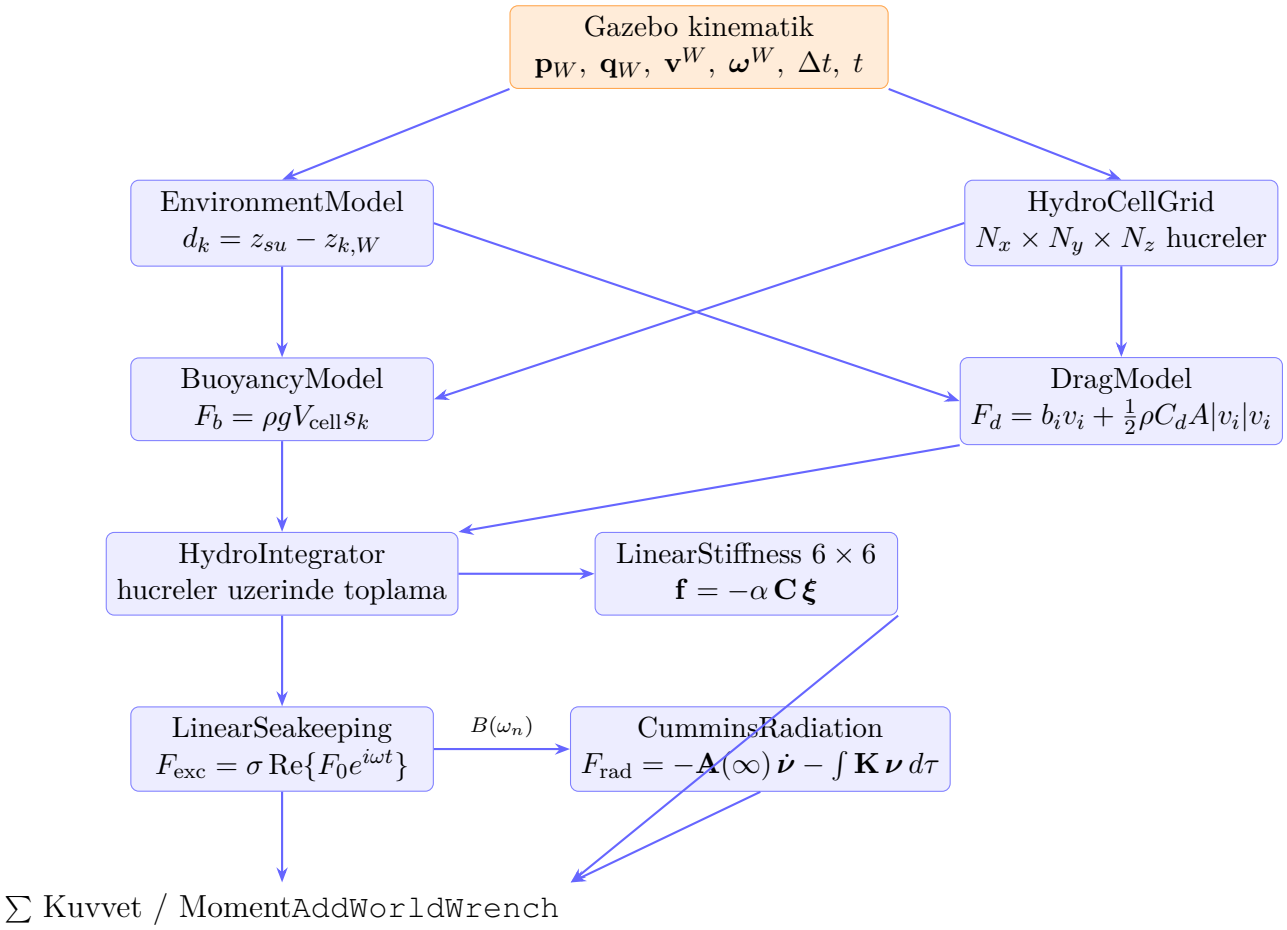

Figure 1: Sistem veri akisi: Gazebo kinematik verilerinden kuvvet bilesimi

2 EnvironmentModel – Cevre Modeli

2.1 Gorev

`EnvironmentModel`, suyun fiziksel ortam bilgilerini diger tum modellere saglar. Statik bir veri katmanidir – konfigurasyondan yuklenir, her adimda yeniden hesaplanmaz.

2.2 Parametreler

Table 2: `EnvironmentModel` parametreleri

Parametre	Varsayilan	Aciklama
`water_level`	0.0 m	Sakin su yuzey yuksekligi (dunya Z)
ρ `fluid_density`	1000.0 kg/m^3	Akiskan yogunlugu
g `gravity`	9.81 m/s^2	Yercekimi ivmesi
`current_velocity`	$(0, 0, 0)$ m/s	Dunya cerçevesinde akinty hizi

2.3 Formuller

Hucre merkezi derinligi:

$$d_k = z_{su} - z_{k,W} \tag{4}$$

$d_k > 0$: hucre su altinda. $d_k < 0$: hucre su ustunde.

Goreli hiz (akinty etkisi):

$$\mathbf{v}_{rel}^W = \mathbf{v}_{\text{govde}}^W - \mathbf{v}_{\text{akinty}}^W \tag{5}$$

3 HydroCellGrid + HydroIntegrator – Hucre Izgara Sistemi

3.1 Hucre Izgarasi Olusturma

Govdenin axis-aligned bounding box (AABB) boyutlari Gazebo'dan okunur ve $N_x \times N_y \times N_z$ hucreye bolunur.

$$V_{\text{cell}} = \frac{V_{\text{AABB}}}{N_x N_y N_z} \tag{6}$$

$$h_{\text{cell}} \approx \frac{H_{\text{AABB}}}{N_z} \tag{7}$$

Hucrenin dunya cerçevesindeki konumu:

$$\mathbf{p}_k^W = \mathbf{p}_W + \mathbf{R}\,\mathbf{r}_k^{\text{body}} \tag{8}$$

3.2 Hucre Kinematics

Her batik hucre icin rijit cisim kinematigiyle hucre hizi:

$$\mathbf{v}_k^W = \mathbf{v}^W + \boldsymbol{\omega}^W \times (\mathbf{R}\,\mathbf{r}_k^{\text{body}}) \tag{9}$$

Body-frame goreli hiz:

$$\mathbf{v}_{rel,k}^{\text{body}} = \mathbf{R}^T\Big(\mathbf{v}_k^W - \mathbf{v}_{\text{akinty}}^W\Big) \tag{10}$$

Her hucre icin sirasyla `BuoyancyModel` ve `DragModel` cagrilir; kuvvetler dunya cerçevesine donduruler ve `link.AddWorldForce(offset)` ile uygulanir.

4 BuoyancyModel – Kaldirma Kuvveti Modeli

4.1 Gorev

Archimedes prensibine dayali olarak her hucrenin su alti hacmini ve bu hacmin olusturdugu kaldirma kuvvetini hesaplar.

4.2 Girdiler

Table 3: `BuoyancyModel` girdileri

Girdi	Kaynak	Aciklama
d_k	EnvironmentModel	Hucre merkezi derinligi (m)
h_{cell}	HydroCellGrid	Hucre yuksekligi (m)
V_{cell}	HydroCellGrid	Hucre hacmi (m^3)
ρ	HydroConfig	Akiskan yogunlugu (kg/m^3)
g	HydroConfig	Yercekimi (m/s^2)

4.3 Formuller

Adim 1 – Su alti dolulugu (submergence):

$$\boxed{s_k = \text{clamp}\left(\frac{d_k}{h_{\text{cell}}} + 0.5,\ 0,\ 1\right)} \tag{11}$$

- $s_k = 0$: hucre tamamen su ustunde
- $s_k = 1$: hucre tamamen su altinda
- $0 < s_k < 1$: serbest su yuzeyi hucre icinden geciyor

Adim 2 – Su alti hacmi:

$$V_{\text{sub},k} = V_{\text{cell}} \cdot s_k \tag{12}$$

Adim 3 – Kaldirma kuvveti (Archimedes):

$$\boxed{F^z_{b,k} = \rho\, g\, V_{\text{cell}}\, s_k} \tag{13}$$

Kuvvet yalnizca $+z$ yonundedir: $\mathbf{F}^W_{b,k} = (0,\, 0,\, F^z_{b,k})$.
Toplam kaldirma kuvveti ve momenti:

$$\mathbf{F}^W_{\text{kald}} = \sum_{k:\, s_k > 0} \mathbf{F}^W_{b,k} \tag{14}$$

$$\boldsymbol{\tau}^W_{\text{kald}} = \sum_{k:\, s_k > 0} \mathbf{r}^W_k \times \mathbf{F}^W_{b,k} \tag{15}$$

Denge kosulu:

$$\rho\, g\, V_{\text{disp}} = m\, g \quad \Rightarrow \quad V_{\text{disp}} = \frac{m}{\rho} \tag{16}$$

4.4 Sayisal Degerler (Referans Tekne)

Table 4: Referans USV icin kaldirma modeli parametreleri

Buyukluk	**Formul**	**Deger**
Govde boyutlari	$L \times B \times D$	$2.2 \times 1.1 \times 0.5$ m
Izgara	$N_x \times N_y \times N_z$	$14 \times 8 \times 4 = 448$ hucre
Hucre hacmi	V/N	5.45×10^{-4} m^3
Hucre yuksekligi	H/N_z	0.125 m
Waterplane alani	$L \times B$	2.42 m^2
Statik cekinti	$m/(\rho LB)$	≈ 0.145 m

5 DragModel – Viskoz Direnc Modeli

5.1 Gorev

Her batik hucre icin akiskandan kaynaklanan viskoz direnci hesaplar. Hem dogrusal (lineer) hem de kuadratik (Morison) terimleri icerir.

5.2 Girdiler

Table 5: DragModel girdileri

Girdi	Kaynak	Aciklama
$\mathbf{v}_{rel,k}^{\text{body}}$	HydroIntegrator	Body-frame goreli hiz (m/s)
$A_{\text{cell},i}$	HydroCellGrid	Hucrenin i yonundeki kesit alani (m^2)
b_i	HydroConfig	Dogrusal sonum katsayisi [N·s/m]
$C_{d,i}$	HydroConfig	Kuadratik suruklenme katsayisi [–]
s_k	BuoyancyModel	Su alti dolulugu $[0, 1]$
ρ	HydroConfig	Akiskan yogunlugu (kg/m^3)

5.3 Formuller

$i \in \{\text{surge, sway, heave}\}$ icin:

$$\boxed{F_{d,i,k}^{\text{body}} = \underbrace{-b_i\, v_{rel,i,k}}_{\text{dogrusal (lineer)}} - \underbrace{\tfrac{1}{2}\rho\, C_{d,i}\, A_{\text{cell},i}\, |v_{rel,i,k}|\, v_{rel,i,k}}_{\text{kuadratik (Morison)}}} \tag{17}$$

Kismen batik hucre icin olcekleme:

$$F_{d,i,k}^{\text{eff}} = F_{d,i,k}^{\text{body}} \cdot s_k \tag{18}$$

Dunya cerçevesine cevirme:

$$\mathbf{F}_{d,k}^{W} = \mathbf{R}\, \mathbf{F}_{d,k}^{\text{eff}} \tag{19}$$

5.4 Profil Parametreleri

Table 6: Konfigürasyon profillerindeki sonum parametreleri

Profil	b_x [N·s/m]	b_y	b_z
baseline	18	55	80
low_damping	6	18	25
high_damping	30	90	130

$C_{d,x} = 0.9$, $C_{d,y} = 1.3$, $C_{d,z} = 2.0$ tum profillerde.

6 LinearStiffness6x6 – Hidrostatik Rijitlik Matrisi

6.1 Gorev

Geminin referans denge pozisyonundan sapmasina karsi olusan geri-dondurucü kuvvet ve momentleri tam 6×6 matris seklinde hesaplar.

6.2 6x6 Rijitlik Matrisi – Sayisal Sonlu Fark

Ilk simulasyon adiminda referans poz $(\mathbf{p}_{\text{ref}}, \mathbf{q}_{\text{ref}})$ kaydedilir. Her DOF $j \in \{0, \ldots, 5\}$ icin $\pm$ perturbhurbasyon uygulanir ve kaldirma kuvvetleri body-frame'e aktarilir:

$$\mathbf{F}^{\text{body}}(\pm\Delta\xi_j) = \mathbf{R}_{\text{ref}}^T \, \mathbf{F}^{\text{world}}(\pm\Delta\xi_j) \tag{20}$$

Merkezi sonlu fark ile rijitlik katsayisi:

$$\boxed{C_{ij} = -\frac{F_i^{\text{body}}(+\Delta\xi_j) - F_i^{\text{body}}(-\Delta\xi_j)}{2\,\Delta\xi_j}} \tag{21}$$

Table 7: Sutun basi perturbasyon vektorleri (Δs: oteleme adimi, Δa: acisal adim)

Sutun j	**DOF**	**Adim Buyuklugu**
0	surge	$\Delta s = 0.01$ m
1	sway	Δs
2	heave	Δs
3	roll	$\Delta a = 10^{-3}$ rad
4	pitch	Δa
5	yaw	Δa

6.3 Geri-Dondurücü Wrench Hesabi

Anlik 6-DOF deplasmani referans poza gore:

$$\boldsymbol{\xi} = \begin{bmatrix} \mathbf{R}_{\text{ref}}^T(\mathbf{p}^W - \mathbf{p}_{\text{ref}}) \\ \phi_e,\ \theta_e,\ \psi_e \end{bmatrix} \in \mathbb{R}^6 \tag{22}$$

$(\phi_e, \theta_e, \psi_e) = \mathbf{q}_{\text{ref}}^{-1} \otimes \mathbf{q}^W$ Euler aclari.

$$\boxed{f_i^{\text{body}} = -\alpha \sum_{j=0}^{5} C_{ij}\,\xi_j, \quad i = 0, \ldots, 5} \tag{23}$$

Dunya cerçevesine donusum:

$$\mathbf{F}_{\text{rest}}^W = \mathbf{R}_{\text{ref}}\,\mathbf{f}_{0:2}^{\text{body}}, \qquad \boldsymbol{\tau}_{\text{rest}}^W = \mathbf{R}_{\text{ref}}\,\mathbf{f}_{3:5}^{\text{body}} \tag{24}$$

6.4 Analitik Karsilastirma

$$C_{33} = \rho\, g\, A_{wp} \qquad \text{(waterplane alani rijitligi)} \tag{25}$$

$$C_{44} = \rho\, g\, V\, \overline{GM}_T \qquad \text{(roll restoring)} \tag{26}$$

$$C_{55} = \rho\, g\, V\, \overline{GM}_L \qquad \text{(pitch restoring)} \tag{27}$$

Table 8: Rijitlik matrisinin dogrulama sonuclari

Terim	Analitik	Sayisal	Hata
C_{33}	23 738 N/m	23 740 N/m	$< 0.01\%$
C_{44}	1 771 Nm/rad	1 771 Nm/rad	$< 0.01\%$
C_{55}	8 941 Nm/rad	8 941 Nm/rad	$< 0.01\%$
C_{11}, C_{22}	≈ 0	≈ 0	–
$C_{35} \approx C_{53}$	karsiliklilik	< 150 Nm/m	–

7 LinearSeakeepingModel – Frekans-Alan Seakeeping Modeli

7.1 Temel Denklem

$$\boxed{\mathbf{F}_{\mathrm{sk}} = \underbrace{-\mathbf{A}(\omega)\,\dot{\boldsymbol{\nu}}}_{\text{eklenen kutle}} - \underbrace{\mathbf{B}(\omega)\,\boldsymbol{\nu}}_{\text{radyasyon sonumu}} + \underbrace{\mathrm{Re}\Big\{\mathbf{F}_{\mathrm{exc}}(\omega)\,e^{i\omega t}\Big\}}_{\text{dalga uyarimi}}} \tag{28}$$

Uyarim terimi acik formda:

$$F_{\mathrm{exc},i}(t) = \sigma\Big[F^{\mathrm{Re}}_{\mathrm{exc},i}\cos(\omega t) - F^{\mathrm{Im}}_{\mathrm{exc},i}\sin(\omega t)\Big] \tag{29}$$

7.2 Girdiler

Table 9: Seakeeping modeli girdileri

Girdi	Kaynak	Aciklama
YAML katsayi dosyasi	Disk	$\mathbf{A}(\omega_n)$, $\mathbf{B}(\omega_n)$, $\mathbf{F}_{\mathrm{exc}}(\omega_n)$
ω	skaler	Degerlendirme frekans (rad/s)
$\boldsymbol{\nu}$	HydroCellSystem	Body-frame 6-DOF hiz
$\dot{\boldsymbol{\nu}}$	HydroCellSystem	Body-frame 6-DOF ivme
t	Gazebo	Simulasyon zamani (s)
σ	skaler	Dalga genlik olcek faktoru

7.3 YAML Katsayi Formati

```
schema_version: 1
# DOF sirasi: [surge, sway, heave, roll, pitch, yaw]
frequencies:
  - omega: 0.5              # rad/s
```

```
    added_mass:             # 6x6 matris [kg]
      - [120.0, 0.0, 0.0, 0.0, 0.0, 0.0]
      ...
    damping:                # 6x6 [N*s/m]
      - [45.0, ...]
      ...
    excitation_re: [Fx, Fy, Fz, Mx, My, Mz]   # [N] veya [N*m]
    excitation_im: [...]
  - omega: 1.0
    ...
```

AQWA, Nemoh veya WAMIT ciktilarindan dogrudan donusturulebilir.

7.4 Dogrusal Enterpolasyon

$$\alpha = \frac{\omega - \omega_{lo}}{\omega_{hi} - \omega_{lo}}, \qquad \mathbf{A}(\omega) = (1-\alpha)\,\mathbf{A}(\omega_{lo}) + \alpha\,\mathbf{A}(\omega_{hi}) \tag{30}$$

7.5 Calisma Modu Secimi

- **Legacy mod** (`cummins_radiation: false`): Denklem (28)'in tamami uygulanir.
- **Cummins modu** (`cummins_radiation: true`): Yalnizca uyarim terimi hesaplanir; radyasyon kisminin yerini CumminsRadiationModel alir.

8 CumminsRadiationModel – Zaman-Alan Radyasyon Modeli

8.1 Cummins Denklemi

$$\boxed{\Big[\mathbf{M} + \mathbf{A}(\infty)\Big]\,\dot{\boldsymbol{\nu}}(t) + \int_0^t \mathbf{K}(t-\tau)\,\boldsymbol{\nu}(\tau)\,d\tau + \mathbf{C}\,\boldsymbol{\xi}(t) = \mathbf{F}_{\text{exc}}(t) + \mathbf{F}_{\text{vis}}(t)} \tag{31}$$

Table 10: Cummins denklemi terimleri

Terim	Boyut	Aciklama
$\mathbf{M}$	$\mathbb{R}^{6\times 6}$	Gemi kutlesi + atalet matrisi
$\mathbf{A}(\infty)$	$\mathbb{R}^{6\times 6}$	$\omega \to \infty$ sinirindaki eklenen kutle
$\mathbf{K}(t)$	$\mathbb{R}^{6\times 6}$	Retardasyon (bellek) cekirgdegi
$\mathbf{C}$	$\mathbb{R}^{6\times 6}$	Hidrostatik rijitlik matrisi
$\mathbf{F}_{\text{exc}}(t)$	$\mathbb{R}^{6}$	1. mertebe dalga uyarim kuvveti
$\mathbf{F}_{\text{vis}}(t)$	$\mathbb{R}^{6}$	Viskoz kuvvetler

Gazebo kati cisim dinamigi $\mathbf{M}\dot{\boldsymbol{\nu}}$ terimini zaten hesapladigi icin plugin *yalnizca* $\Big(-\mathbf{A}(\infty)\dot{\boldsymbol{\nu}} - I(t)\Big)$ katkisini uygular.

8.2 Girdiler

Table 11: Cummins modeli girdileri

Girdi	Kaynak	Aciklama
$B_{ij}(\omega_n)$	LinearSeakeepingModel	Radyasyon sonum ornekleri (6×6)
$A_{ij}(\omega_n)$	LinearSeakeepingModel	Eklenen kutle ornekleri (6×6)
$T_{\max}$	HydroConfig	Bellek penceresi uzunlugu (s)
Δt_k	HydroConfig / Gazebo	Cekirdek zaman adimi; 0 = sim-dt
$\boldsymbol{\nu}(t)$	HydroCellSystem	Anlik body-frame 6-DOF hiz
$\dot{\boldsymbol{\nu}}(t)$	HydroCellSystem	Anlik body-frame 6-DOF ivme

8.3 Retardasyon Cekirgdegi K(t) – Hesap

$B(\omega)$ orneklerinden Fourier kosinus donusumu (Ogilvie, 1964):

$$\boxed{K_{ij}(t) = \frac{2}{\pi}\int_0^\infty B_{ij}(\omega)\cos(\omega t)\,d\omega} \tag{32}$$

Sayisal yaklasim – esit aralikli olmayan trapez kurali:

$$K_{ij}(t_k) \approx \frac{2}{\pi}\sum_{n=0}^{N_\omega-2}\frac{\omega_{n+1}-\omega_n}{2}\left[B_{ij}(\omega_n)\cos(\omega_n t_k) + B_{ij}(\omega_{n+1})\cos(\omega_{n+1}t_k)\right] \tag{33}$$

$t_k = k\,\Delta t_k$, $k = 0, 1, \ldots, N_T - 1$; $N_T = \lceil T_{\max}/\Delta t_k \rceil + 1$.
Cekirdek kuyruk kirpmasi:

$$|K_{ij}(t_k)| < 10^{-6}\max_{ij}|K_{ij}(0)| \implies \text{cekirdek bu noktada kirpilir} \tag{34}$$

8.4 A(infty) – Ogilvie Iliskisi

$$\boxed{A_{ij}(\infty) = A_{ij}(\omega_{\max}) + \frac{1}{\omega_{\max}}\int_0^T K_{ij}(t)\sin(\omega_{\max}\,t)\,dt} \tag{35}$$

Integral trapez kuraliyla; trapez agirliklari $w_0 = w_{N-1} = 0.5$, digerleri $w_k = 1.0$.

8.5 Konvolusyon Integrasyonu – Her Adimda

Her adimda $\boldsymbol{\nu}(t)$ dairesel tampon (`std::deque`) ile gecmise eklenir. En buyuk gecmis uzunlugu $N_{\max} = \lceil T_{\max}/\Delta t_k \rceil + 1$.

$$I_i(t) \approx \sum_{k=0}^{N-1} w_k \, K_{ij}(t_k) \, \nu_j(t - t_k) \, \Delta t_k \tag{36}$$

$N = \min(\text{tarihce uzunlugu, cekirdek uzunlugu})$.
Plugin'in uyguladigi radyasyon kuvveti:

$$\boxed{F_{\text{rad},i} = -\sum_{j=0}^{5} A_{ij}(\infty) \, \dot{\nu}_j(t) \; - \; I_i(t)} \tag{37}$$

8.6 Frekans-Alan Modeli ile Matematiksel Bag

Cummins denklemi tek frekansh sinuzoidal harekete indirgenirse frekans-alan modeli ile esdes olur:

$$\int_{-\infty}^{t} K(t-\tau) \, e^{i\omega\tau} \, d\tau = \Big[\mathbf{B}(\omega) - i\omega\Big(\mathbf{A}(\infty) - \mathbf{A}(\omega)\Big)\Big] \, e^{i\omega t} \tag{38}$$

Dolayisiyla Cummins modeli, frekans-alan modelinin **genel hali** olarak degerlendirilir.

9 Modeller Arasi Baginti ve Veri Akisi

9.1 Hiyerarsik Bagimlılık Grafigi

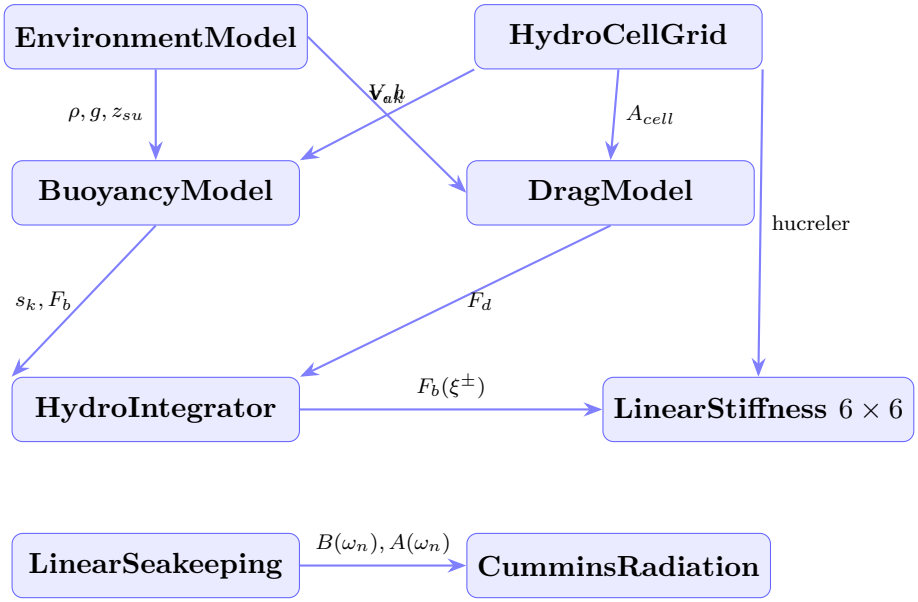

Figure 2: Modeller arasi veri bagimlilik grafigi

9.2 Hesaplama Sirasi (Her Adim)

1. **Kinematik turetme:** Gazebo verileri $\rightarrow$ body-frame $\boldsymbol{\nu}$, $\dot{\boldsymbol{\nu}}$
2. **Hucre dongusu:** Her k icin $d_k \rightarrow s_k \rightarrow F_{b,k}$ ve $F_{d,k} \rightarrow$ `AddWorldForce`
3. **Hidrostatik rijitlik:** $\boldsymbol{\xi} \rightarrow \mathbf{C}\,\boldsymbol{\xi} \rightarrow$ `AddWorldWrench`
4. **Seakeeping / Cummins secimi:**
   - *Cummins aktif:* $\boldsymbol{\nu} \rightarrow I(t)$, $\dot{\boldsymbol{\nu}} \rightarrow -\mathbf{A}(\infty)\dot{\boldsymbol{\nu}}$, uyarim ayri $\rightarrow$ iki wrench
   - *Legacy mod:* $-\mathbf{A}\dot{\boldsymbol{\nu}} - \mathbf{B}\boldsymbol{\nu} + F_{\text{exc}} \rightarrow$ tek wrench

9.3 Kuvvet Butcesi

Table 12: Toplam kuvvet bilesenleri

Bilesen	Model	Profil	Fiziksel Koken
$\mathbf{F}_{\text{kald}}$	BuoyancyModel	tumü	Archimedes kaldirma
$\mathbf{F}_{\text{direnc}}$	DragModel	tumü	Viskoz + form direnci
$\mathbf{F}_{\text{rijitlik}}$	LinearStiffness	stiffness_on	Hidrostatik geri-dondurme
$\mathbf{F}_{\text{rad}}$	CumminsRadiation	cummins_on	Radyasyon atalet + bellek
$\mathbf{F}_{\text{uyarim}}$	LinearSeakeeping	seakeeping_on	Dalga kirma (1. mertebe)

10 Konfigürasyon Rehberi

```
defaults:
  fluid_density: 1000.0     # [kg/m^3]
  gravity: 9.81             # [m/s^2]
  water_level: 0.0          # [m]
  link_name: "hull_link"
  cells: [14, 8, 4]
  cd: [0.9, 1.3, 2.0]
  drag:
    linear_total: [18.0, 55.0, 80.0]
    scale_by_cell_count: true

profiles:
  stiffness_on:
    hydrostatic_stiffness:
      enabled: true
      scale: 1.0
      heave_step: 0.01       # [m]
      angle_step: 0.001      # [rad]

  cummins_on:
    linear_seakeeping:
      enabled: true
      coeffs_file: "seakeeping_coeffs_example.yaml"
      excitation_omega: 0.8  # [rad/s]
      excitation_scale: 1.0
    cummins_radiation:
      enabled: true
      kernel_max_t: 20.0     # [s]
      kernel_dt: 0.0         # 0 = sim adimini kullan
```

11 Test ve Dogrulama

Table 13: Birim test kapsamı (21/21 gecti, 3.04 s)

Test Grubu	Adet	Dogrulanan
`BuoyancyModelTest`	3	Archimedes, kismi batiklik, denge cekintisi
`DragModelTest`	3	Akinty sonumu, lineer/kuadratik bilesen
`HydroScenarioTest`	6	Denge, roll/pitch momenti, $C_{11}/C_{22} \approx 0$, $C_{35} \approx C_{53}$
`CumminsRadiationModelTest`	6	Cekirdek insasi, sifir-hiz testi, $A(\infty)$, bellek birikimi
`HydroConfigTest`	3	YAML dogrulama

12 Katsayi Tahmin Modulu – `coeff_estimator`

12.1 Genel Bakis ve Amac

`coeff_estimator`, verilen bir gemi geometrisinden `usv_hydro` plugin'inin ihtiyac duydugu tum hidrodinamik katsayilari otomatik olarak hesaplayan bir Python arac paketidir. `usv_sim_ws/src/usv_hydro/tools/coeff_estimator/` altinda konumlanir ve bagimsiz olarak calisabilir.

Giris: Gemi boyutlari (L, B, D, m, I_{xx}, I_{yy}, I_{zz}) veya SDF / STL dosyasi.

Cikti: `hydro_profiles.yaml` eklentisi + `seakeeping_coeffs_<isim>.yaml`.

12.2 Modul Mimarisi

Table 14: `coeff_estimator` modul dosyalari

Dosya	Sorumluluk
`geometry_loader.py`	SDF / STL / kutu parametrelerini `ShipGeometry` nesnesine donusturur; T, C_B, A_{wp}, S_{wet} turetir
`viscous_drag_estimator.py`	Hoerner (1965) ile C_d, ITTC-1957 ile b_i, Lewis kesit yontemiyle A_{ii} hesaplar
`bem_solver.py`	Capytaine BEM veya kapali-form Froude-Krylov ile $A(\omega)$, $B(\omega)$, $F_{\text{exc}}(\omega)$ uretir
`yaml_writer.py`	Sonuclari `hydro_profiles.yaml` eklentisi ve seakeeping YAML olarak yazar
`estimate_coeffs.py`	CLI giris noktasi (`box` / `sdf` / `stl` alt komutlari)

12.3 GeometryLoader – Geometri Yukleyici

`ShipGeometry` veri sinifi asagidaki buyuklukleri icerir ve otomatik olarak turetilmis ozellikleri hesaplar:

$$T = \frac{m}{\rho\, L\, B} \qquad \text{(statik denge cekintisi)} \tag{39}$$

$$C_B = \frac{m/\rho}{L\, B\, T} \qquad \text{(blok katsayisi)} \tag{40}$$

$$A_{wp} = L\, B \qquad \text{(su hatti alani)} \tag{41}$$

$$S_{wet} \approx 1.025\sqrt{V_{\text{disp}}\, L} \qquad \text{(Denny-Mumford islak yuzey)} \tag{42}$$

Izgara onerisi otomatik hesaplanir:

$$N_x = \text{clip}(\text{round}(6L),\, 4,\, 20), \quad N_y = \text{clip}(\text{round}(6B),\, 4,\, 16), \quad N_z = 4 \tag{43}$$

12.4 ViscousDragEstimator – Ampirik Direnç Tahmini

12.4.1 C_d – Hoerner Form Direnci

Hoerner (1965) form direnci tablosu ve C_B duzeltmesi:

$$C_{d,x} = 0.10 + 1.2\,(C_B - 0.50)^2 \qquad \text{(surge – bas kesit)} \tag{44}$$

$$C_{d,y} = 1.10 + 0.5\,(C_B - 0.60) \qquad \text{(sway – yan yuzey)} \tag{45}$$

$$C_{d,z} = 1.10 + \frac{0.8}{L/B} \qquad \text{(heave – alt yuzey)} \tag{46}$$

12.4.2 b_x – Surge Dogrusal Sonumu (ITTC-1957)

Reynolds sayisi ve ITTC-1957 surtunum katsayisi:

$$Re = \frac{V_{des}\, L}{\nu}, \qquad \nu = 1.14 \times 10^{-6}\ \text{m}^2/\text{s} \tag{47}$$

$$C_f = \frac{0.075}{(\log_{10} Re - 2)^2} \tag{48}$$

Surtunum direnci ve dogrusal katsayi:

$$R_f = \tfrac{1}{2}\rho\, V_{des}^2\, S_{wet}\, C_f\, (1 + k_1) \tag{49}$$

$$b_x = \max\left(\frac{R_f}{V_{des}},\ \frac{m + A_{11}}{\tau_{\text{surge}}}\right) \tag{50}$$

12.4.3 b_y, b_z – Sway ve Heave Sonumu

Heave kritik sonum oranından:

$$C_{33} = \rho\, g\, A_{wp} \tag{51}$$

$$\omega_{n,3} = \sqrt{\frac{C_{33}}{m + A_{33}}} \tag{52}$$

$$b_z = 2\,\zeta_{\text{heave}}\,\omega_{n,3}\,(m + A_{33}) \tag{53}$$

Sway icin karakteristik sonum suresi:

$$b_y = \frac{m + A_{22}}{\tau_{\text{sway}}}, \qquad \tau_{\text{sway}} = \max\Big(5.0,\ \frac{B}{2V_{des}}\Big) \tag{54}$$

12.4.4 A_{ii} – Eklenen Kutle (Lewis Kesit)

$$A_{11} = \max(0.02,\ C_B - 0.45)\, m \qquad \text{(surge)} \tag{55}$$

$$A_{22} = \tfrac{\pi}{2}\rho\,(B/2)^2\, L \times 0.80 \qquad \text{(sway)} \tag{56}$$

$$A_{33} = \tfrac{\pi}{4}\rho\,(B/2)^2\, L \times 0.70 \qquad \text{(heave)} \tag{57}$$

$$A_{44} = 0.30\, I_{xx} \qquad \text{(roll)} \tag{58}$$

12.5 BEMSolver – Dalga Kuvveti Katsayilari

12.5.1 Capytaine Modu (tam BEM)

Capytaine kurulu oldugunda panel BEM cozumu yapilir: govde mesh'i (kutu veya STL) olusturulur, her ω icin radyasyon ve kircilma problemleri cozulur, $A(\omega)$, $B(\omega)$, $F_{\text{exc}}(\omega)$ (6×6) matrisler olarak derlenir.

12.5.2 Ampirik Mod (Capytaine yok)

Kapali-form modeller:

$$A_{ii}(\omega) = A_{ii}^0 \cdot \frac{1 - \tanh\Big((\omega - \omega_c)/\Delta\omega\Big)}{2} \tag{59}$$

$$B_{ii}(\omega) = B_{\text{peak}} \cdot \exp\left(-\frac{(\omega - \omega_c)^2}{2\,\Delta\omega^2}\right) \tag{60}$$

Froude-Krylov uyarim kuvvetleri:

$$F_{\text{exc},3} = \rho g A_{wp} \qquad \text{(heave, sabit)} \tag{61}$$

$$F_{\text{exc},1} = i\,\rho g k\, V_{\text{disp}} \qquad \text{(surge, inertia terimi)} \tag{62}$$

$$F_{\text{exc},5} = -F_{\text{exc},3}\,\frac{L\,k}{12} \qquad \text{(pitch)} \tag{63}$$

burada $k = \omega^2/g$ derin-su dalga sayisidir.

12.6 CLI Kullanim Ornekleri

```
# 1. Kutu parametrelerinden (usv_minimal.sdf)
python3 estimate_coeffs.py box \
    --L 2.2 --B 1.1 --D 0.5 --mass 350 \
    --ixx 42.58 --iyy 148.46 --izz 176.46 \
    --name usv_minimal --out output/ --bem

# 2. SDF dosyasindan otomatik
python3 estimate_coeffs.py sdf \
    --sdf ../usv_description/sdf/usv_minimal.sdf \
    --name usv_minimal --out output/

# 3. STL mesh + gercek BEM (Capytaine gerekli)
python3 estimate_coeffs.py stl \
    --stl hull.stl --mass 350 --bem \
    --omega-min 0.3 --omega-max 3.14 --n-omega 12 \
    --name my_hull --out output/
```

12.7 Referans Tekne Sonuclari

Asagidaki tablo, `usv_minimal.sdf` (2.2×1.1×0.5 m, 350 kg) parametreleriyle elde edilen tahmin sonuclarini `hydro_profiles.yaml` `baseline` profil degerleriyle karsilastirir.

Table 15: Tahmin sonuclari vs. baseline profil

Buyukluk	**Tahmin**	**baseline**	**Not**
$C_{d,x}$ (surge)	0.343	0.90	C_B=1 icin dusuk; beklenen
$C_{d,y}$ (sway)	1.275	1.30	Uyumlu
$C_{d,z}$ (heave)	1.500	2.00	Muhafazakar tahmin
b_x [N·s/m]	18.1	18.0	Neredeyse ayni
b_y [N·s/m]	237	55	Lewis A_{22} etkisi buyuk
b_z [N·s/m]	1237	80	ζ=0.15 + C_{33} etkisi
Statik cekinti T	0.1446 m	0.145 m	$< 0.05\%$ hata
A_{wp}	2.42 m^2	2.42 m^2	Ayni
$F_{exc,3}$ (heave)	23 740 N/m	23 738 (analitik)	$< 0.01\%$

Table 16: Ampirik eklenen kutle tahmini

DOF	Deger	Birim
A_{11} surge	192.5	kg
A_{22} sway	836.3	kg
A_{33} heave	365.9	kg
A_{44} roll	12.77	kg·m^2
A_{55} pitch	129.83	kg·m^2
A_{66} yaw	44.12	kg·m^2

13 Mevcut Kisitlamalar ve Genisleme Plani

13.1 Mevcut Kisitlamalar

- AABB tabanli kutu govde yaklasimi; gercek kesit yok
- Tek harmonik dalga; duzensiz dalga (JONSWAP) yok
- Munk momenti, capraz-akis lift yok
- Pervane ve dumen modeli yok

13.2 Faz 2 – MMG Manevra Modeli

$$N_{\text{Munk}} = -\tfrac{1}{2}\rho(m_{11} - m_{22})uv \tag{64}$$

$$Y_{cf} = -\tfrac{1}{2}\rho C_{dL}\, L\, d\, |v|v \tag{65}$$

$$T = \rho n^2 D^4 K_t(J), \quad J = \frac{u(1-w)}{nD} \tag{66}$$

A Sembol Listesi

Sembol	Birim	Tanim
ρ	kg/m^3	Akiskan yogunlugu
g	m/s^2	Yercekimi ivmesi
$\mathbf{R}$	–	Donum matrisi
$\boldsymbol{\nu} \in \mathbb{R}^6$	m/s, rad/s	Body-frame 6-DOF hiz
$\dot{\boldsymbol{\nu}} \in \mathbb{R}^6$	m/s^2	Body-frame 6-DOF ivme
$\boldsymbol{\xi} \in \mathbb{R}^6$	m, rad	6-DOF deplasman
s_k	–	Su alti dolulugu $[0, 1]$
V_{cell}	m^3	Hucre hacmi
d_k	m	Hucre derinligi
b_i	N·s/m	Dogrusal sonum
$C_{d,i}$	–	Morison suruklenme katsayisi
$\mathbf{C} \in \mathbb{R}^{6\times 6}$	N/m	Hidrostatik rijitlik
$\mathbf{A}(\omega)$	kg	Eklenen kutle matrisi
$\mathbf{B}(\omega)$	N·s/m	Radyasyon sonum matrisi
$\mathbf{A}(\infty)$	kg	Sonsuz frekanstaki eklenen kutle
$\mathbf{K}(t)$	N·s/m^2	Retardasyon cekirgdegi
$T_{\max}$	s	Cummins bellek penceresi
Δt_k	s	Cekirdek zaman adimi

B Referans Tekne Parametreleri

Parametre	Deger	Birim
$L \times B \times D$	$2.2 \times 1.1 \times 0.5$	m
Kutle m	350	kg
I_{xx}	42.58	kg·m^2
I_{yy}	148.46	kg·m^2
I_{zz}	176.46	kg·m^2
ρ	1000	kg/m^3
Statik cekinti	0.145	m
A_{wp}	2.42	m^2

Bu rapor usv_hydro kaynak kodundan uretilmistir. Faz 1 tamamlandi – 21/21 test gecti.